

Cloud protocol

Distributed Data Processing Environments

This handout holds the setup, the conventions, and the cleanup that every cloud session assumes.

Setup Once, before the first cloud session:

1. Log in your GMail account in the browser.
2. Click on “Redeem Now” in the voucher email, or enter information here: <https://console.cloud.google.com/edu>
3. Go to Cloud Console,¹ create a project, and select the education credits billing account.
4. Confirm that no payment method is attached. When the credit runs out, everything stops.
5. Navigate to Compute Engine > VM Instances and select “Enable”. Wait until complete.
6. Install the Google Cloud CLI.² On Windows, prefer installing it inside WSL2.
7. Log in with `gcloud auth login`, then select your project with `gcloud config set project ...`
8. Run `gcloud auth application-default login` as well. Tools and libraries authenticate through this, and not through the previous one.

Cloud Console The browser interface at <https://console.cloud.google.com>.

- Check the project selector at the top. Everything you do applies to the project named there.
- A path such as Compute Engine > VM Instances means: open the navigation menu on the left, select the first, and then the second. The search box at the top reaches the same pages.
- The terminal icon on the top right opens Cloud Shell: a shell with `gcloud` already installed and logged in, at no cost, and with a home directory that survives between sessions.
- On the VM Instances page, the SSH button of an instance opens a session in the browser, with nothing installed on your machine.

Guardrails Once, and before creating anything:

- Billing > Budgets & caps: create a budget of 50 USD for your project, alerting at 50%, 90%, and 100%.

Conventions

- Region *us-central1* and zone *us-central1-a*, unless a guide says otherwise. They are among the cheapest.
- Label every resource with `course=adpd`, so that you can find what you left behind.

¹<https://console.cloud.google.com>

²<https://cloud.google.com/sdk/docs/install>

Connecting

- Connect with: `gcloud compute ssh name --zone us-central1-a`
- Add `--tunnel-through-iap` for an instance that has no external address.
- Copy files with `gcloud compute scp`.

At the end of every session

1. Destroy what you created, with `tofu destroy` or from the console.
2. Confirm that Compute Engine > VM Instances is empty.
3. Confirm that Compute Engine > Disks is empty. Disks outlive the instances they belonged to.
4. Confirm that Cloud Storage > Buckets holds nothing you still pay for.